Topic 3.5 – Hazardous Earth

"Every year, natural disasters affect one in thirty people on Earth. Whether developing or industrialised, all nations are at risk, and the field of natural hazards is today one of the fastest growing areas of research in the Earth and Climate Sciences." – University College London, Earth Sciences Department

Movement of the Earth's land masses, from Pangaea to present day are evidence that forces beneath our feet are at work. Seismic and volcanic activity creates hazards as populations have grown and inhabited more of the Earth. Although hazardous, earthquakes and volcanoes create new landforms and can support life on Earth from flora and fauna to populations.

As technology has evolved, the capacity to predict and mitigate against tectonic hazard events has improved although the impact of an event can leave communities and countries devastated. Risks from tectonic hazards varies spatially and over time, with continued research and development there may be a point in the future when it will be possible to mitigate against the vulnerability to risk.

Currently there are a number of strategies which help the international community, governments and individuals cope with the risks associated with tectonic hazards however there are varying global levels of resilience and ability to adapt to the risks presented.

1. What is the evidence for continental drift and plate tectonics?	
Key Ideas	**Content**
1.a. There is a variety of evidence for the theories of continental drift and plate tectonics.	• Theories of continental drift and plate tectonics including: ◦ the basic structure of the Earth including the lithosphere, asthenosphere and the role of convection currents ◦ evidence for sea-floor spreading; paleomagnetism; the age of sea floor rocks ◦ evidence from ancient glaciations ◦ fossil records.
1.b. There are distinctive features and processes at plate boundaries.	• Earth's crustal features and processes, including: ◦ the global pattern of plates and plate boundaries ◦ the features and processes associated with divergent (constructive) plate boundaries ◦ the features and processes associated with convergent plate boundaries including oceanic-continental, oceanic-oceanic (destructive) and continental-continental (collision) boundaries ◦ the features and processes associated with conservative plate boundaries.

2. What are the main hazards generated by volcanic activity?	
Key Ideas	**Content**
2.a. There is a variety of volcanic activity and resultant landforms and landscapes.	• Different types of volcanoes to investigate their causes and features including: ◦ explosive eruptions (higher viscosity magma) located at convergent (destructive) plate boundaries ◦ effusive eruptions (lower viscosity magma) and landforms located at divergent (constructive) plate boundaries ◦ eruptions not at plate boundaries (hot spots) such as the Hawaiian chain and the East African Rift Valley ◦ size and shape of different types of volcanoes, including super-volcanoes ◦ the volcanic explosive index (VEI) measure of assessing volcanic activity.
2.b. Volcanic eruptions generate distinctive hazards.	• Different types of volcanic eruptions and the different types of hazards they generate including: ◦ lava flows, pyroclastic flows, gas emissions, tephra and ash ◦ lahars and flooding associated with the melting of ice ◦ tsunamis associated with explosive eruption.

3. What are the main hazards generated by seismic activity?	
Key Ideas	**Content**
3.a. There is a variety of earthquake activity and resultant landforms and landscapes.	• Earthquake characteristics to investigate their causes and features including: ◦ shallow-focus earthquakes ◦ deep-focus earthquakes ◦ the different measures of assessing earthquake magnitude (Richter, moment magnitude scale, modified Mercalli intensity scale) ◦ the effects earthquakes have on landforms and landscapes including the development of escarpments and rift valleys.
3.b. Earthquakes generate distinctive hazards.	• Hazards generated by earthquakes, including: ◦ ground shaking and ground displacement ◦ liquefaction ◦ landslides and avalanches ◦ tsunamis associated with sea-bed uplift and underwater landslides ◦ flooding.

4. What are the implications of living in tectonically active locations?	
Key Ideas	**Content**
4.a. There are a range of impacts people experience as a result of volcanic eruptions.	• **Case studies** of **two** countries at contrasting levels of economic development to illustrate: ◦ reasons why people choose to live in tectonically active locations ◦ the impacts people experience as a result of volcanic eruptions ◦ economic, environmental and political impacts on the country.
4.b. There are a range of impacts people experience as a result of earthquake activity.	• **Case studies** of **two** countries at contrasting levels of economic development to illustrate: ◦ reasons why people choose to live in tectonically active locations ◦ the impacts people experience as a result of earthquake activity ◦ economic, environmental and political impacts on the country.

5. What measures are available to help people cope with living in tectonically active locations?	
Key Ideas	**Content**
5.a. There are various strategies to manage hazards from volcanic activity.	• **Case studies** of **two** countries at contrasting levels of economic development to illustrate strategies used to cope with volcanic activity including: ◦ attempts to mitigate against the event, such as lava diversion channels ◦ attempts to mitigate against vulnerability such as community preparedness ◦ attempts to mitigate against losses, such as rescue and emergency relief.
5.b. There are various strategies to manage hazards from earthquakes.	• **Case studies** of **two** countries at contrasting levels of economic development to illustrate strategies used to cope with hazards from earthquakes including: ◦ attempts to mitigate against the event such as land-use zoning ◦ attempts to mitigate against vulnerability such as building design ◦ attempts to mitigate against losses such as insurance.
5.c. The exposure of people to risks and their ability to cope with tectonic hazards changes over time.	• How and why have the risks from tectonic hazards changed over time including: ◦ changes in the frequency and impacts of tectonic hazards over time ◦ the degree of risk posed by a hazard and the probability of the hazard event occurring (the disaster risk equation) ◦ possible future strategies to cope with risks from tectonic hazards. • The relationship between disaster and response including the Park model.